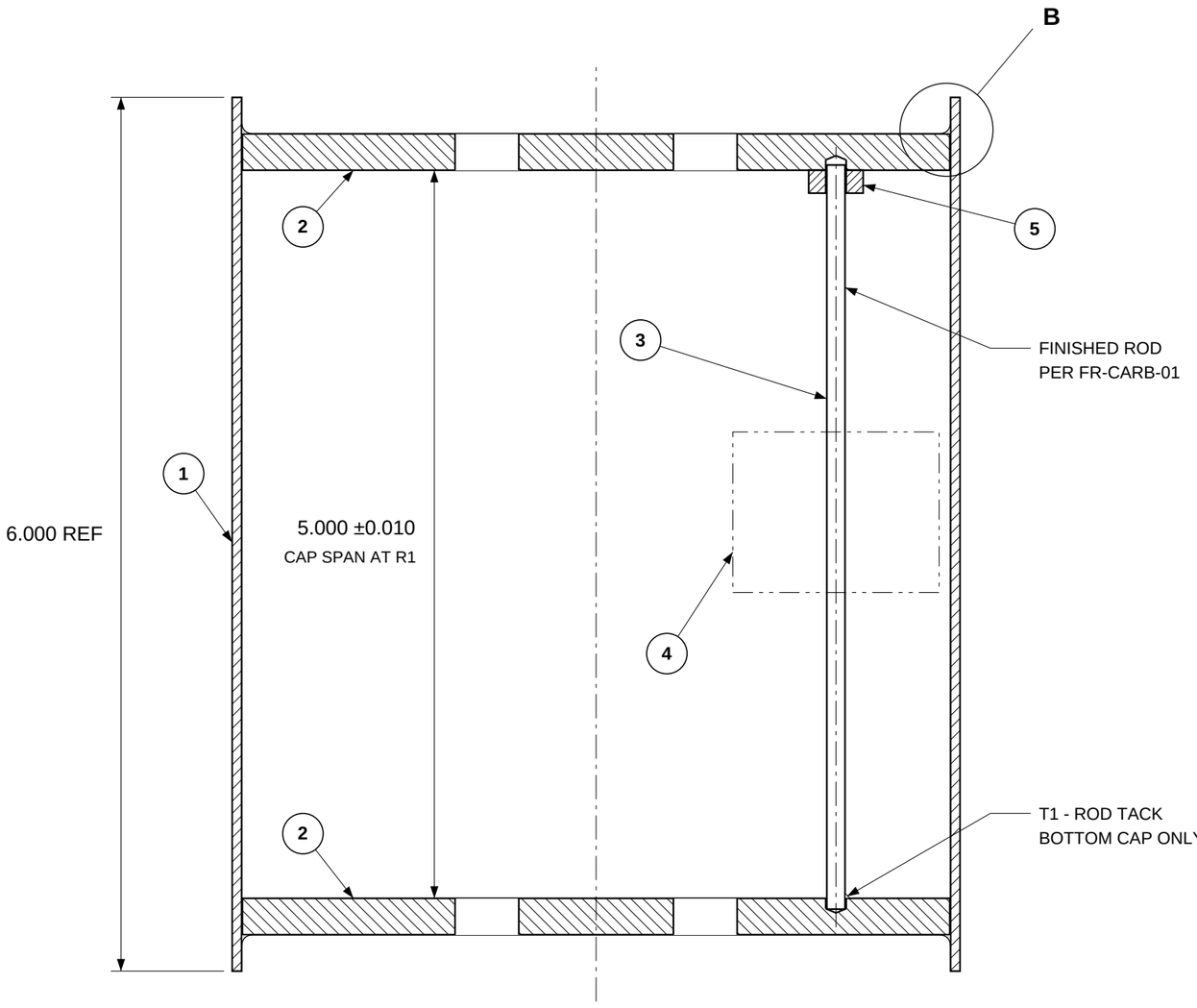
SECTION A-A

LOOKING +X; RIGHT IS -Y - SCALE 1:1

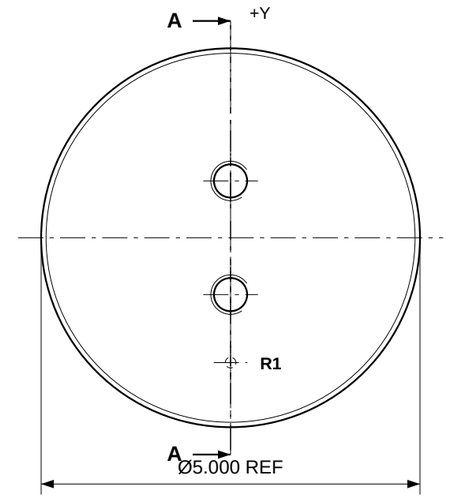


FIXED-LENGTH ROD SEATS IN BOTTOM REGISTER; TOP REGISTER RETAINS CLEARANCE.

PORTS AND ROD SHARE THIS PLANE. FLOAT ENVELOPE IS REFERENCE ONLY.

TOP OUTSIDE END VIEW

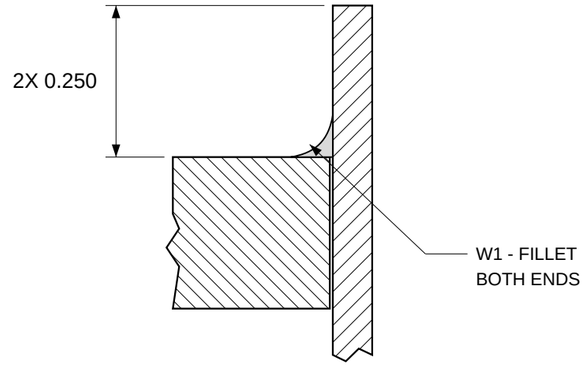
LOOKING DOWN -Z - SCALE 1:2



BOTH R1 REGISTERS ON -Y; BOTH PORT PAIRS ON ±Y.

DETAIL B - CLOSURE JOINT

SCALE 4:1 - LOWER END MIRRORED



WELD PROFILE DIAGRAMMATIC; SIZE NOT ESTABLISHED.

CAP OD / BORE SLIP GAP SHOWN; DO NOT CHAMFER WELD ROOT.

PARTS AT CLOSURE

ITEM	QTY	MATERIAL	DESCRIPTION / DEFINITION
1	1	HSM-0008	316 SS TUBE; Ø5.000 x 0.065 WALL x 6.000
2	2	HSM-0009	316 SS FINISHED END CAP; END-CAP-01
3	1	HSM-0336	316 SS FLOAT GUIDE ROD; FR-CARB-01
4	1	HSM-0032	ASA AERO MAGNETIC FLOAT; INSTALL BEFORE CLOSURE
5	1	316 SS	UPPER FLOAT STOP; DEFINITION IN NOTE 5

ASSEMBLY REQUIREMENTS

1. Fabrication state of HSM-0004 before permanent fittings. Component ports, registers and edge preparation: END-CAP-01. Stock dimensions are nominal.

2. Finished cap inside-face spacing AT ROD AXIS: 5.000 ±0.010. Set both outside faces 0.250 below adjacent tube rims. Align both port pairs on ±Y; both R1 register axes coaxial on -Y, opposite the reservoirs' +Y sensing stations.

3. Use the fixed-length rod per FR-CARB-01. Seat and tack it in the bottom register before the bottom closure weld. Install the float before top closure; verify top engagement, free travel and full cap seating at the specified spacing.

4. W1: continuous internal-corner fillet around BOTH cap perimeters. ER316L .030 filler. Clean bore band and cap outside face; preserve the square cap weld root. T1: tack rod to bottom cap. Welding, inspection, hydro and passivation: pressure-vessel procedure, steps 2-8.

5. Upper stop: Ø0.374 stock, Ø9/64 THRU, 0.1575 +0.0000/-0.0079 long; square faces, deburr. T2: tack concentrically over top R1 before closure. Keep bore clear. With inlet fitted, retain 0.039 MIN axial air above the stopped float; verify through full radial guide play. Stop attachment and installed clearance require qualification.

RELEASE ITEMS: W1 SIZE AND WELD QUALIFICATION; REMAINING UNSPECIFIED ASSEMBLY TOLERANCES.

HSM-0004 / FABRICATION STATE

DIMENSIONS GOVERN.

DO NOT SCALE.

PRINT AT 100%

FOR STATED SCALES.

REF = REFERENCE

HOME SODA MACHINE

CARBONATOR VESSEL - CLOSURE ASSEMBLY

PROTOTYPE

FOR REVIEW

MATERIAL	UNITS	QUANTITY
SEE PARTS LIST	**INCHES**	**1 PER MACHINE**

DRAWING	REV	DATE	SCALE	SHEET
CV-WLD-01	**C**	**2026-10-28**	**AS SHOWN**	**1 OF 1**